imperfection and it increases the electrical conductivity of a semiconductor material such as silicon. The ability of ferromagnetic material such as iron, nickel etc., to be magnetized and demagnetized depends on the presence of imperfections. Crystal defects are classified as follows

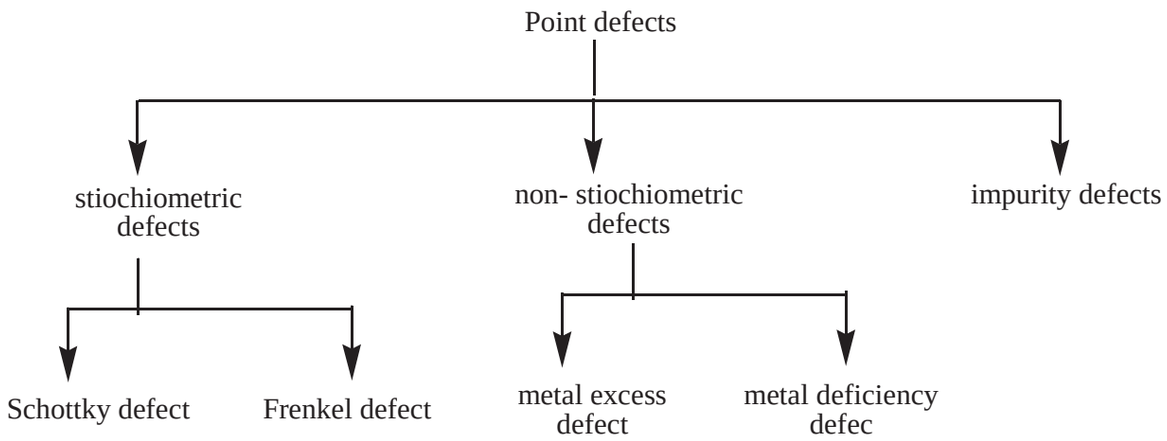
1) Point defects

2) Line defects

3) Interfacial defects

4) Volume defects

In this portion, we concentrate on point defects, more specifically in ionic solids. Point defects are further classified as follows



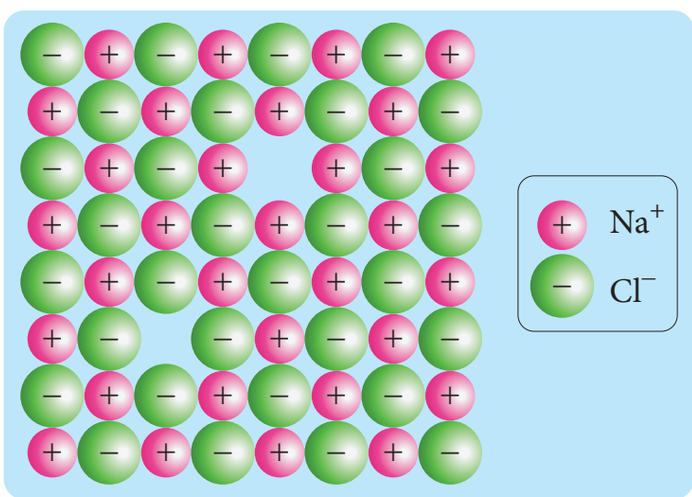
Stoichiometric defects in ionic solid:

This defect is also called intrinsic (or) thermodynamic defect. In stoichiometric ionic crystals, a vacancy of one ion must always be associated with either by the absence of another oppositely charged ion (or) the presence of same charged ion in the interstitial position so as to maintain the electrical neutrality.

6.7.1 Schottky defect:

Schottky defect arises due to the missing of equal number of cations and anions from the crystal lattice. This effect does not change the stoichiometry of the crystal. Ionic solids in which the cation and anion are of almost of similar size show schottky defect. Example: NaCl.

Presence of large number of schottky defects in a crystal, lowers



Schottky Defect

XII U6 Solid State - Jerald.indd 193 3/29/2019 6:30:31 PM